Name of Technology: **"Silver Screen printed RFID tags technology for NFC applications**

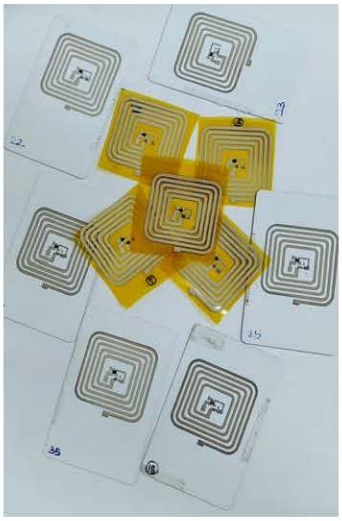
There has been an increasing need and rapid progress in the field of wireless identification and communications. Nowadays, RFID technology is widely used in modern lives. Radio frequency identification (RFID) technology provides compelling means for the automatic identification and tracking of items. This is achieved by labelling them with battery free remotely addressable electronic tags composed of an antenna and integrated circuit (IC). The use of propagating electromagnetic waves at ultra-high frequencies (UHF) for powering and communicating with the passive tags enables rapid interrogation of large quantity of tags through various media from the long distance. In comparison to bar-codes, RFID tags allow the data stored in them to be updated wirelessly at any time. This helps people deal with several problems with a less consuming of both financial and human resources. With significant advantages such as requiring no batteries or maintenance, contactless identification, perfect tolerance of a grubby environment, a high performance in security with password in a chip that cannot be duplicated and a longer life of equipment's than that of barcodes and magnetic strip cards, RFID is used widely in various fields. For instance, currently RFID technology is mainly focuses on the following aspects, such as payment by mobile phones, transportation payments, asset management and retail sales, product tracking, transportation and logistics, animal identification, inventory systems, libraries and so on.

Passive RFID communication typically occurs at 13.56 MHz, where at low power levels the bulk of the energy radiated by the reader is contained in the near-field (distance from some cm to about 30 cm) and such systems use loop antennas in the readers and the tags.

C-MET has developed NFC RFID Tags printing technology using low temperature (up to 150^{0}C depending on substrate) curable paste. Presently, these RFID NFC tags are imported from the handful of suppliers having worldwide monopoly. These tags are successfully tested for near field Communications at 13.56 MHz

C-MET has filed an Indian patent application (Application No. 20241103396 dated 24.02.2024) related to this technology.

13.56 MHz Frequency RFID TAGS



Specifications of 13.56 MHz RFID Tag

Parameters	In-house developed NFC Flex Stamp  (NTAG 213F/216F) Screen printed using silver paste	Commercial NFC Flex stamp  (Pulse's own W9908/W9909 C-clips) Copper Etched
Frequency [MHz]	13.56	13.56
Reading Distance [mm]	10	10
Impedance (Ω)	50	50
Inductance (µH)	1	0.9
Resistance (Ω)	30-40	1.55
Q Factor	3	49
Operating Temperature [°C]	+24 To +48	-40 To +85